FEWER CRASHES, DEADLIER ROADS

Where Seattle Should Act to Reach Vision Zero

IE 6600: Project Report 1 (Data Story)

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Percentage of Effort Contributed by Student: 100%

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Executive Summary

Seattle's Vision Zero program aims to end traffic deaths and serious injuries on city streets by 2030 [1]. Using the city's collision records [2], I analyzed more than 262,000 reported crashes to understand where, when, and why the most serious crashes happen, and where the Seattle Department of Transportation (SDOT) should focus its limited safety budget.

The central finding is a paradox: Seattle has fewer crashes than before Vision Zero began, but severe crashes have risen by about a third. The analysis shows that these crashes are concentrated around three factors: speeding and drivers not stopping for pedestrians, late-night hours on clear, dry roads, and a small number of locations where pedestrians and cyclists are seriously hurt, led by Aurora Ave N.

Based on these findings, I recommend that SDOT target speeding, prioritize safety upgrades for people walking and biking at the highest-risk locations, focus enforcement on the evening rush and late night, and measure progress by severe crashes rather than total crashes.

I used Python to clean and prepare the data and to build precisely designed charts, and Tableau to map high-risk locations and explore patterns quickly, combining both tools into a single data story.

1. Business Understanding

1.1 Background and business problem

In 2015, the City of Seattle adopted Vision Zero, a commitment to eliminate traffic deaths and serious injuries on city streets by 2030 [1]. The program is managed by SDOT, which designs streets, sets speed limits, installs signals and crosswalks, and runs public safety campaigns, working alongside the Seattle Police Department on enforcement.

With the 2030 deadline approaching, the problem is that the most serious crashes are not declining. From 2015 to 2025, 283 people were killed on Seattle streets, about 26 per year, and severe collisions in 2021 to 2025 averaged about 34% more per year than in 2010 to 2014. At the same time, SDOT's safety budget is limited: it cannot redesign every street, patrol every corridor, or run campaigns everywhere at once. The business problem is therefore one of prioritization: where, when, and on which behaviors should SDOT focus to prevent the most severe crashes?

1.2 Stakeholders

- **Primary: SDOT's Vision Zero team,** which decides where to invest in street redesigns, signals, lighting, and campaigns. This analysis is designed to support their decisions.
- **Seattle Police Department,** which carries out traffic enforcement such as speed and impairment patrols.
- **The public,** especially pedestrians and cyclists, who face the highest risk of serious harm.
- **City leadership and council,** who approve safety budgets and monitor Vision Zero progress.

1.3 How analytics and predictive models can help

This project is descriptive: it identifies patterns in historical crashes. These patterns are also the foundation for predictive models that SDOT could use next. For example, a classification model could estimate the probability that a crash will be severe based on time, light, weather, location, and recorded behaviors, and a location-level risk model could rank corridors by expected severe crashes. Such models would let SDOT rank projects by expected lives saved per dollar and anticipate emerging hotspots rather than reacting after crashes occur.

1.4 Organizational capacity

SDOT is well positioned to use these outputs. It already collects and publishes collision data with precise coordinates through its open data portal [2], maintains GIS capability, and its Vision Zero plan explicitly commits to data-driven safety investment [1]. The main constraints are budget and data quality: some fields, such as contributing factors, are only recorded when an officer notes them.

1.5 Business goal and success criteria

Business goal: Reduce fatal and serious-injury traffic collisions in Seattle by focusing limited safety resources on the locations, times, and behaviors that drive severe crashes.

Success criteria for this project:

- Identify the trend in severe collisions and whether Seattle is on track for Vision Zero.
- Identify the driver's behaviors, times, conditions, and locations most associated with severe crashes.
- Translate the findings into at least three specific, actionable recommendations for SDOT.
- Long term (for SDOT): a sustained year-over-year decline in severe collisions, measured separately from total collisions.

2. Data Description & Summary

2.1 Dataset overview

The data comes from SDOT Collisions - All Years, published by the City of Seattle through Data.gov and the Seattle GIS open data portal [2], with field definitions from the city's metadata [3]. Each row is one reported collision. The raw file contained 262,998 records and 64 variables, covering October 2003 to October 2026. Variables describe when and where the crash happened, how severe it was, who was involved (people, vehicles, pedestrians, cyclists), road, weather, and light conditions, and contributing factors such as speeding and impairment.

I used full calendar years 2004 to 2025 for the long-term trend (259,677 collisions) and the most recent ten years, 2016 to 2025 (92,581 collisions, 2,233 severe), for the detailed analysis, since recent years best reflect today's streets and driving patterns.

2.2 Preprocessing

I cleaned the data in Python (pandas) in Google Colab. Table 1 summarizes each step.

Step	What I did	Why
Duplicates	Removed 3 duplicate report numbers	Avoid double-counting crashes
Missing severity	Removed 380 rows with no severity (all had 0 injuries)	Cannot classify severity
Dates and times	Converted date text to datetime; created Year, Month, Day of Week, Weekend, Hour	Enable trend and time-of-day analysis
Missing times	Flagged 23% of rows with a date but no time (Hour left blank, rows kept)	Avoid guessing a time
Severity flag	Created Severe = 1 for serious injury or fatal	Matches the Vision Zero target
Behavior flags	Standardized Y/N/1/0 values; blank = not recorded	Consistent analysis
Conditions	Grouped detailed weather, road, and light values into 5–6 categories	Reliable percentages per group
Coordinates	Converted Washington State Plane (feet) to latitude/longitude	Required for mapping in Tableau

Step	What I did	Why
Columns	Dropped 21 internal or mostly empty fields (64 → 43)	Focus on analysis-ready fields

Table 1. Preprocessing steps. Final dataset: 262,615 rows × 43 columns.

2.3 Missing values and data quality

Figure 1 shows the share of records missing or unknown for key fields. Severity, location, and coordinates are nearly complete. About 20% of records have unknown weather, road, or light conditions, mostly from citizen-filed reports, which rarely include details. The speeding and inattention flags appear mostly blank because they only record a "Y" when an officer notes the factor; a blank means "not recorded," not "no." To limit this bias, the behavior analysis uses complete police reports only.

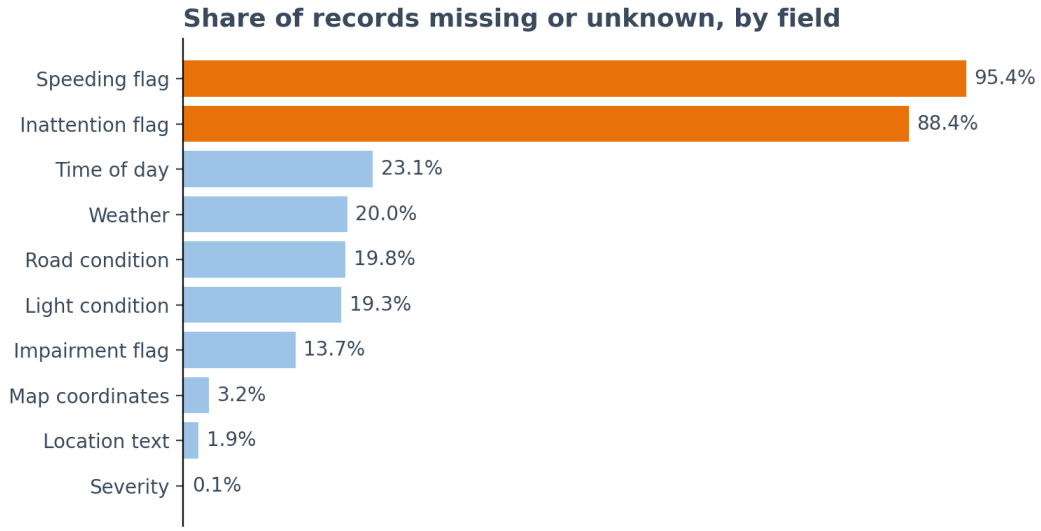


Figure 1. Missing or unknown values by field (raw data). Orange fields only record "Y" when noted by an officer.

2.4 Distributions and key statistics

Severity is highly imbalanced (Figure 2): 81.2% of collisions caused property damage only, 17.0% caused injuries, 1.7% caused serious injuries, and 0.2% were fatal. Severe crashes are therefore rare, about 1.9% of all collisions, which is why this project focuses on the share and count of severe crashes rather than total crashes.

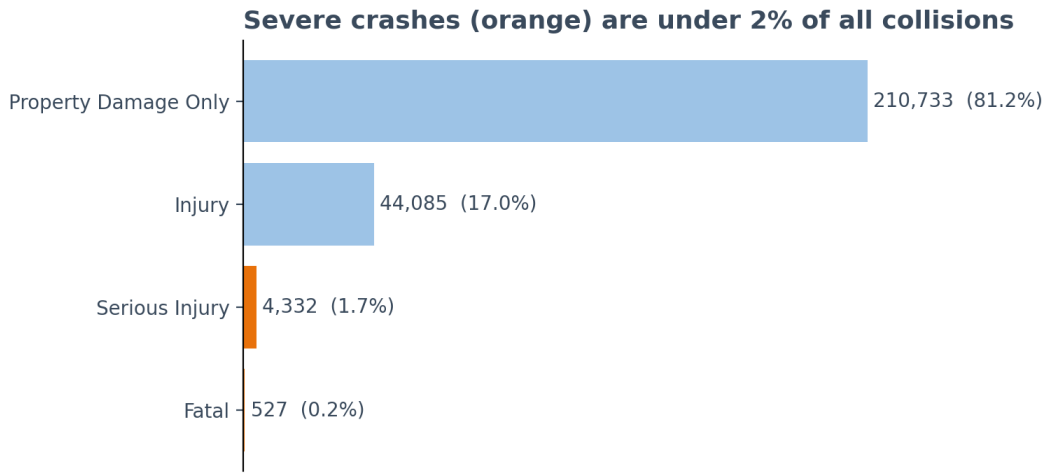


Figure 2. Collisions by severity, 2004–2025.

Table 2 summarizes key numeric variables. A typical collision involves two people and two vehicles; pedestrian and cyclist involvement is rare per crash. Annual totals vary widely because collisions fell sharply after 2019, while severe collisions per year stayed comparatively stable.

Variable	Mean	Median	Std. dev.	Max
People involved per crash	2.15	2	1.50	93
Vehicles involved per crash	1.69	2	0.86	15
Pedestrians per crash	0.04	0	0.21	6
Cyclists per crash	0.03	0	0.17	3
Collisions per year	11,804	12,609	3,213	16,016
Severe collisions per year	221	209	43	327

Table 2. Descriptive statistics, full years 2004–2025.

Other distributions: 88% of reports were filed by police and 12% by citizens; 64% of collisions occurred mid-block and 34% at intersections; and the most common collision types were hitting a parked car (21%), angle crashes (16%), and rear-ends (15%).

2.5 Relationships between variables

Because Severe is a binary outcome, I examined relationships by comparing the share of crashes that are severe across groups, which is more meaningful than raw counts. The strongest relationships found were:

- **Pedestrian involvement:** 18.6% of crashes involving a pedestrian were severe, compared with 1.3% of vehicle-only crashes (about 14 times higher). Pedestrian count also had the highest correlation with Severe among the numeric fields (r = 0.20).
- **Speeding:** 7.3% of crashes with speeding recorded were severe, compared with 2.2% without.
- **Time of day:** the severe share ranged from 1.4% at 8 AM to 5.6% at 2 AM.

- **Light and weather:** crashes after dark were more likely to be severe than daytime crashes, while rain was associated with lower severity.

These relationships informed the business questions in Section 3.

3. Identifying Business Questions

I framed the data story around five business questions, each tied to a decision SDOT controls: where to spend, when to enforce, which behaviors to target, and how to message campaigns.

#	Business question	Decision it informs
1	Is Seattle on track to meet its Vision Zero goal?	Whether current strategy is working; which metric to track
2	Which driver behaviors should SDOT target with enforcement and campaigns?	Enforcement focus and campaign content
3	When should SDOT focus enforcement and safety campaigns?	Scheduling of patrols and speed checks
4	Should campaigns focus on bad-weather days or clear nights?	Timing and messaging of public campaigns
5	Which locations are pedestrians and cyclists most often seriously hurt?	Where to invest in crosswalks, signals, lighting, and bike lanes

Table 3. Business questions and the decisions they support.

4. Answering Business Questions

The visualizations below form a step-by-step narrative: first the overall problem (trend), then why severe crashes happen (behaviors), when (time and conditions), and where (locations). Across all charts, orange highlights the key finding, and other data is shown in soft blue or gray, following the decluttering and attention principles from the course [4]. Each chart uses an action title that states the findings.

4.1 Is Seattle on track to meet its Vision Zero goal?

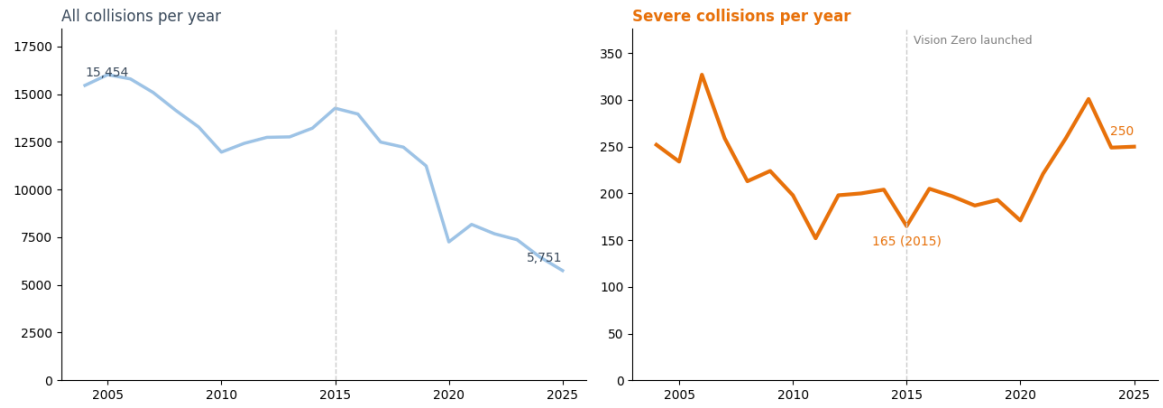


Figure 3. Crashes have fallen by nearly half, but severe crashes are rising.

Why this visualization: Two side-by-side line charts show change over time clearly. Separate panels let each series keep its own scale; on a single axis, severe crashes (hundreds per year) would look flat next to total crashes (thousands).

What it shows: All collisions fell from 15,454 in 2004 to 5,751 in 2025. Severe collisions reached their lowest point in 2015 (165), the year Vision Zero launched, and have risen since, reaching 250 in 2025. Comparing five-year averages to avoid cherry-picking single years, total collisions fell about 44% (about 12,600 to 7,100 per year), while severe collisions rose about 34% (about 190 to 256 per year).

Business meaning: Seattle is not on track. Falling total crashes can make progress look better than it is, so SDOT should track severe crashes as its primary success metric. Part of the drop in total crashes may reflect fewer minor crashes being reported, but serious crashes are almost always reported, so their rise is real.

4.2 Which driver behaviors should SDOT target?

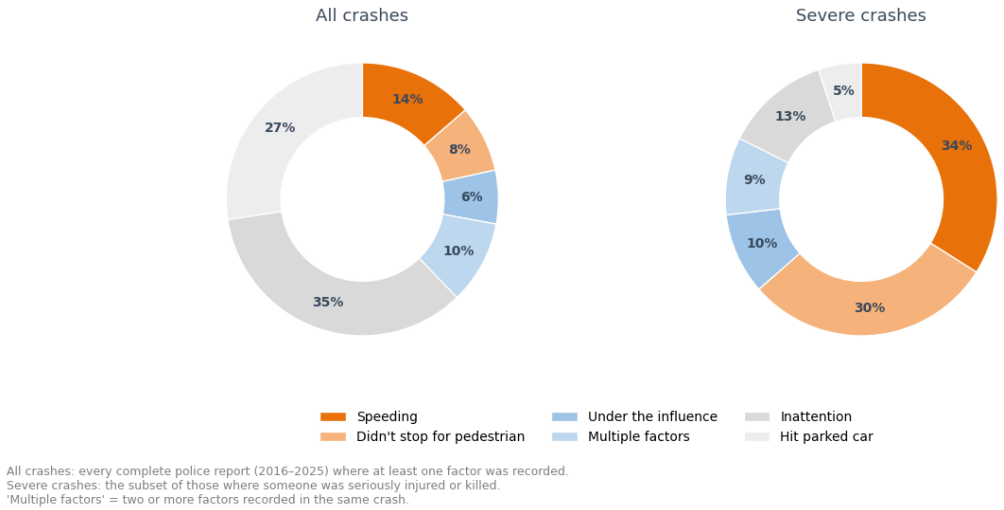


Figure 4. Speeding and failing to stop for pedestrians dominate severe crashes.

Why this visualization: Two donut charts show part-to-whole composition, and placing them side by side makes the shift in the mix immediately visible. Each crash was assigned to exactly one category (or "Multiple factors") so the slices sum to 100%. Only complete police reports were used to reduce bias from missing data.

What it shows: In all crashes with a recorded factor, the most common factors were inattention (35%) and hitting a parked car (27%), typically low-speed crashes. In severe crashes, the picture flips: speeding (34%) and drivers not stopping for pedestrians (30%) together account for 64%. Among fatal crashes with a recorded factor, speeding alone accounts for 54%.

Business meaning: The most common crash causes are not the deadliest. Distracted-driving campaigns reduce fender-benders, but speed and failure to yield to pedestrians are what turn crashes deadly, so enforcement and design should prioritize them.

4.3 When should SDOT focus enforcement and campaigns?

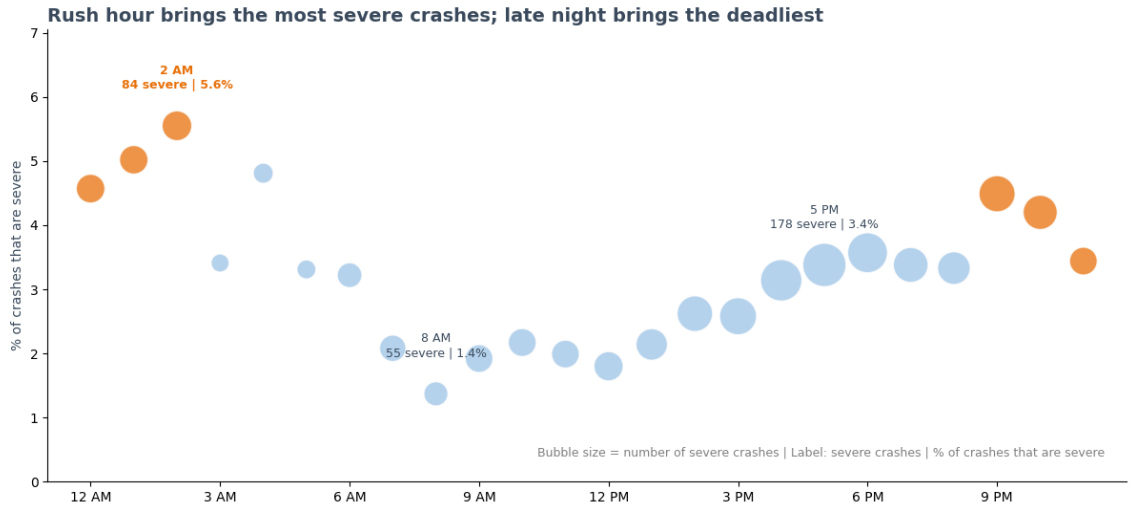


Figure 5. Rush hour brings the most severe crashes; late night brings the deadliest.

Why this visualization: A bubble chart shows two measures at once: the height shows risk (share of crashes that are severe) and the bubble size shows volume (number of severe crashes). This reveals that the hours with the most severe crashes are not the hours where each crash is most dangerous. It uses the roughly 80% of crashes with a recorded time.

What it shows: The evening rush has the largest number of severe crashes, peaking at 178 at 5 PM, because roads are busiest. Late at night, there are fewer crashes, but each is far more dangerous: 5.6% of crashes at 2 AM were severe, compared with 1.4% at 8 AM, about four times higher.

Business meaning: SDOT and police should focus on two windows: the evening rush (3–9 PM) to reach the most incidents, and late night (9 PM–3 AM), when risk per crash is highest and speeding and impairment are more likely.

4.4 Should campaigns focus on bad-weather days or clear nights?

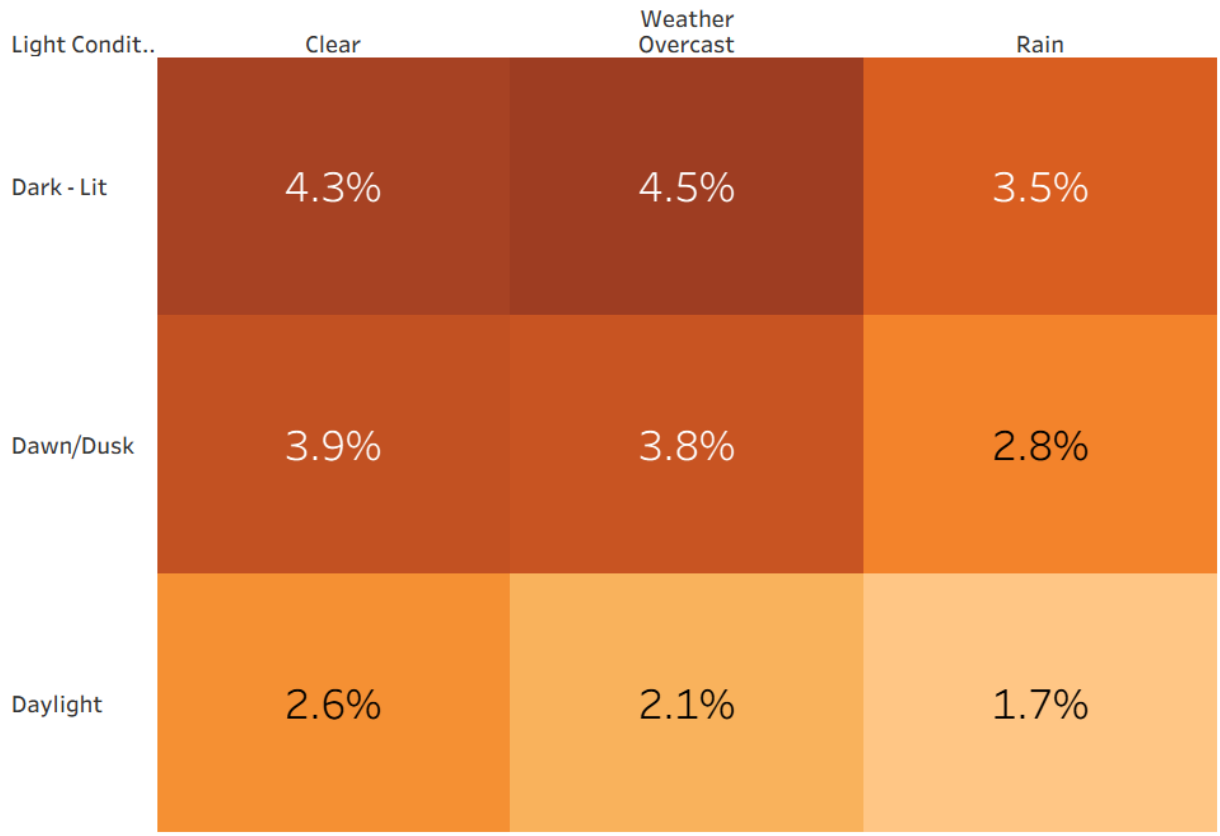


Figure 6. A crash on a clear night is 2.5 times as likely to be severe as one on a rainy day.

Why this visualization: A highlight table compares two categorical variables at once, with color intensity showing risk, so the most and least dangerous combinations stand out instantly. Cells with fewer than 100 crashes were excluded so that small samples do not produce misleading colors.

What it shows: Crashes after dark were the most likely to be severe: 4.3% on clear nights and 4.5% on overcast nights. Rainy daytime crashes were the least severe, at 1.7%. In every light condition, rain had the lowest severity.

Business meaning: Contrary to common assumptions, bad weather is not the main danger. Drivers likely slow down in the rain, while dry, open roads at night invite speeding. Campaigns should shift from "drive carefully in the rain" toward "slow down on clear nights," which aligns with the speeding finding.

4.5 Where are pedestrians and cyclists most often seriously hurt?

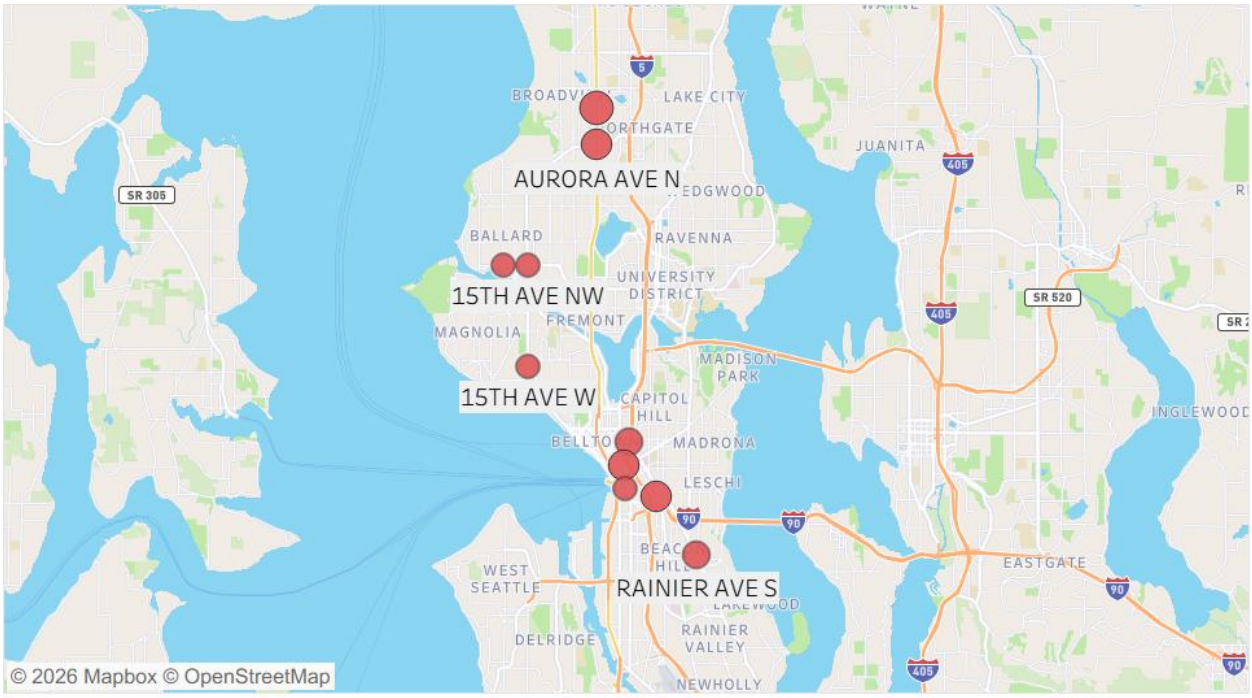


Figure 7. Where pedestrians and cyclists are most often seriously hurt, led by Aurora Ave N.

Why this visualization: A street map is the most intuitive way to show location, since decision-makers immediately recognize the streets. Limiting the map to the top locations, sized by number of severe crashes, keeps it uncluttered and focused on where action is needed.

What it shows: Pedestrians and cyclists were involved in fewer than 8% of crashes but 1,089 of 2,233 severe crashes (about 49%). The top locations were Aurora Ave N between N 117th Pl and N 125th St (6 severe crashes), Aurora Ave N near N Northgate Way (5), 5th Ave & Spring St (5), 12th Ave S & S Weller St (5), Boren Ave & Pine St (4), and Rainier Ave S (4).

Business meaning: Protecting people walking and biking addresses nearly half of the severe-crash problem. Aurora Ave N holds two of the top four locations, making it the clearest first priority for pedestrian safety investment.

5. Business Recommendations

Based on the data story, I recommend five actions for SDOT, ordered by expected impact on deaths and serious injuries.

1. **Target speeding.** Expand speed cameras, lower speed limits on high-risk arterials, and use street designs that naturally slow traffic, especially at night. Evidence: speeding is involved in 34% of severe and 54% of fatal crashes with a recorded factor (Figure 4).
2. **Protect people walking and biking at the highest-risk locations.** Upgrade crosswalks, add leading pedestrian intervals (giving pedestrians a head start before vehicles get a green light), improve

lighting, and add protected bike lanes, starting on Aurora Ave N. Evidence: pedestrians and cyclists account for about half of severe crashes, concentrated at a few locations (Figure 7).

3. **Focus enforcement on two time windows.** Schedule patrols and speed checks for the evening rush (3–9 PM) and late night (9 PM–3 AM). Evidence: the most severe crashes occur at rush hour, while the highest risk per crash occurs late at night (Figure 5).
4. **Shift campaign messaging.** Move from "drive carefully in the rain" to "slow down on clear nights." Evidence: clear-night crashes are about 2.5 times as likely to be severe as rainy-day crashes (Figure 6).
5. **Track severe crashes, not total crashes.** Report severe collisions as the primary Vision Zero metric, separately from total collisions. Evidence: total crashes fell about 44% while severe crashes rose about 34% (Figure 3).

Limitations and Next Steps

- **Recorded factors are incomplete.** Speeding, inattention, and impairment are only recorded when an officer notes them, so these behaviors are likely undercounted. The findings show association, not proof of cause.
- **Missing time and conditions.** About 23% of crashes have no recorded time and about 20% have unknown conditions; these were excluded only from the analyses that required those fields.
- **Exposure is not measured.** Busy streets naturally have more crashes. Adding traffic volume or street length would allow crash rates per vehicle or per mile.
- **Recent reporting changes.** The sharp drop in total collisions after 2019 may partly reflect the pandemic and changes in reporting.
- **Next step: predictive modeling.** A classification model predicting crash severity, and a corridor-level risk model, would help SDOT rank projects by expected impact.

Reflection: Working with Python and Tableau

I used both Python and Tableau in this project and found that each tool had clear strengths, so I chose the tool based on the task.

What worked well in Python

- **Data cleaning at scale:** Python (pandas) handled the 263,000-row dataset efficiently, including removing duplicates, parsing dates and times, grouping condition categories, and converting the city's State Plane coordinates into latitude and longitude for mapping.
- **Custom calculations:** Python made it straightforward to build analysis-specific measures, such as the Severe flag, five-year averages for a fair trend comparison, and assigning each crash to exactly one contributing-factor category so the donut charts summed to 100%.

- **Full control over design:** with matplotlib, I could precisely apply the course's decluttering principles: removing borders and gridlines, labeling data directly, using action titles, and highlighting only the key finding in orange.
- **Reproducibility:** every step is saved in the notebook, so the full analysis can be rerun from the raw data.

What worked well in Tableau

- **Mapping:** once the coordinates were converted, Tableau turned latitude and longitude into an interactive Seattle street map in minutes, with tooltips showing the street name and number of severe crashes.
- **Fast exploration:** dragging fields onto rows, columns, and color made it quick to test different views, such as the light and weather highlight table, before settling on the final design.
- **Calculated fields and filters:** features like a % Severe calculated field and Top N filters made it easy to focus each chart on what mattered.

Challenges and lessons learned

- **Filter order matters in Tableau:** my Top 10 locations map initially showed only 5 points, because Tableau applied the Top N filter before the other filters. Adding those filters to context fixed it, and taught me to check how a tool processes filters.
- **Small samples can mislead:** in the highlight table, the darkest cells were based on as few as 12 crashes. Excluding cells with fewer than 100 crashes made the chart honest and readable.
- **Consistency across tools:** using the same color palette (orange for the key finding, soft blue for context) in both Python and Tableau made the charts feel like one coherent story.
- **Choosing the right tool:** Python was best for heavy data preparation and precisely designed charts, while Tableau was best for maps and quick interactive exploration. Using both together produced a stronger data story than either alone.

References

[1] City of Seattle, “Vision Zero: Seattle’s plan to end traffic deaths and serious injuries by 2030,” Seattle, WA, USA, 2015. [Online]. Available: https://www.seattle.gov/Documents/Departments/beSuperSafe/VisionZeroPlan.pdf

[2] Seattle Department of Transportation, “SDOT collisions – All years,” Data.gov, dataset. [Online]. Available: https://catalog.data.gov/dataset/sdot-collisions-all-years. Accessed: Oct. 26, 2026.

[3] Seattle Department of Transportation, “Collisions: Open data metadata,” City of Seattle, data dictionary. [Online]. Available: https://www.seattle.gov/Documents/Departments/SDOT/GIS/Collisions_OD.pdf. Accessed: Oct. 26, 2026.

[4] A. Chen, “Lessons in storytelling; Clutter is your enemy; Focus on the audience’s attention,” lecture slides, IE 6600, Northeastern Univ., Seattle, WA, USA, 2026.

Appendix: Variables Used in the Analysis

Group	Variables
Time	Date, Year, Hour, Has_Time
Severity	Severity, Severe, Fatalities
Road users	Pedestrian_Involved, Cyclist_Involved, Vulnerable_User_Involved
Contributing factors	Speeding, Under_Influence, Inattention, Ped_ROW_Not_Granted, Hit_Parked_Car
Conditions	Weather, Light_Condition
Location	Location, Latitude, Longitude
Data quality / identifiers	Report_Source, Incident_Key

Table 4. Variables used (22 of 43 cleaned columns). Tools: Python (pandas, matplotlib) in Google Colab; Tableau Public.